

Ventrogluteal intramuscular administration of testosterone undecanoate 1000 mg/4 mL

Standard operating procedure and training guide

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Section 1: Standard operating procedure

Purpose and scope

To ensure the safe, consistent administration of testosterone undecanoate 1000 mg/4 mL (Reandron 1000) by deep intramuscular injection into the gluteal muscle, with the ventrogluteal site preferred, minimising the risk of injury and of pulmonary oil microembolism (POME) [1][4][5][6]. Applies to credentialed clinicians — medical practitioners, nurse practitioners and registered nurses — who administer this product in the facility.

Responsibilities

- **Prescribing clinician:** confirms the indication, the dose schedule and informed consent [1].
- **Administering clinician:** must be trained and assessed as competent in ventrogluteal intramuscular technique and in recognising and managing POME and anaphylaxis (see Section 2). Competency is recorded on the staff file [5].
- **Clinic manager:** ensures emergency equipment (oxygen, adrenaline, monitoring, a means of calling 000) is available and in date, and keeps the current Product Information on file [1][8].

Pre-procedure checks

- Verify patient identity, indication, allergies and informed consent, including any previous reaction to an oil-based injection [1].
- Confirm the patient is well enough to remain for the observation period.
- Confirm the product, dose (1000 mg in 4 mL), expiry and batch number. Check the Product Information for contraindications and precautions [1][2].

Equipment

- testosterone undecanoate 1000 mg/4 mL (Reandron 1000), one 4 mL vial or ampoule [1].
- 18G needle for drawing up — the solution is a viscous oil [5].
- 20–22G needle for injection [10]; this procedure specifies 21G × 38 mm, or 21G × 50 mm where the BMI is over 30, so that the needle reaches muscle [5].
- 5 mL syringe, gloves, antiseptic swabs, gauze, adhesive dressing, sharps container [5].
- Emergency kit: oxygen, adrenaline for intramuscular use, monitoring equipment, a means of calling 000 [8].

Procedure

1. **Position the patient** side-lying with the upper hip exposed and relaxed, or supine with slight hip flexion. Ensure comfort and privacy [5].
2. **Landmark the ventrogluteal site** using the hand opposite to the hip (left hand for the right hip): place the heel of the hand over the greater trochanter, point the index finger towards the anterior superior iliac spine and spread the middle finger back along the iliac crest. The centre of the V between the two fingers, over gluteus medius, is the injection point [4][5].
3. **Prepare the site** with antiseptic and allow it to dry. Put on gloves [5].
4. **Draw up** the full 4 mL with the 18G needle, then change to the injection needle and expel air [1][5].
5. **Insert the needle at 90°** to the skin, to a depth that places the tip in muscle, not subcutaneous tissue [5]. Aspirate for blood return [5].
6. **Inject very slowly**, over approximately 2 minutes [10]. The Australian Product Information requires very slow injection without giving a time [1][3]; the manufacturer's administration guidance for the same formulation specifies about 2 minutes [10]. Slow injection reduces the pressure that forces oil into vessels [6][7].
7. **Withdraw the needle** and apply gentle pressure with gauze. Do not massage the site. Apply a dressing and dispose of sharps immediately [5].

Observation and management

The Product Information requires the patient to be observed during and immediately after each injection [1][3]. This procedure requires observation for **at least 15 minutes**. Pulmonary oil microembolism presents within minutes as cough, breathlessness, malaise, sweating, chest pain, dizziness, paraesthesia or syncope [1][3]; it is uncommon — a reported rate below 0.1% per injection in post-marketing data [6] — but can be frightening and occasionally serious [7]. Anaphylaxis has also been reported [1].

- **Suspected POME:** stop, reassure, position the patient comfortably (sitting if breathless), give high-flow oxygen, monitor observations, and call 000 if symptoms are severe or do not settle within minutes [6][7]. Most episodes resolve with supportive care [7].
- **Suspected anaphylaxis:** lay the patient flat (legs slightly raised), do not let them stand, give adrenaline 1:1000 intramuscularly into the outer mid-thigh — 0.01 mg/kg up to 0.5 mg (0.5 mL) per dose for an adult — call 000, and repeat the dose after 5 minutes if there is no improvement. Observe for at least 4 hours after the last dose, in hospital. Follow the current ASCIA guideline [8].
- Document the event, notify the prescriber, complete the adverse event form (Appendix C) and report to the TGA [9].

Documentation

Record the date and time, product, batch number and expiry, dose, side injected (for example “right ventrogluteal”), needle gauge and length, the administering clinician, any reaction, the patient's condition at the end of the observation period, and the next due date. Maintenance injections are usually given every 10 to 14 weeks; the interval between the first and second injection may be shortened to a minimum of 6 weeks on the prescriber's instruction [1][3].

Reporting and escalation

Any serious or unexpected reaction, especially POME or anaphylaxis, is escalated immediately to the on-call medical officer, recorded in the local incident system and reported to the TGA [9]. The event is reviewed, and this procedure or the training updated if needed.

Review

Reviewed annually, or whenever the Product Information, national guidance or local policy changes.

Section 2: Training guide

Learning objectives

On completion the clinician can:

- explain why the gluteal muscle, and the ventrogluteal site in particular, is used for this large-volume oily injection [1][4][5];
- landmark the ventrogluteal site accurately [4][5];
- choose needle gauge and length for the patient's build [5];
- inject 4 mL of oily solution deep into muscle, over approximately 2 minutes [1][3][10];
- recognise and manage POME and anaphylaxis [6][7][8];
- document and report correctly [9].

Rationale

Reandron 1000 is licensed for deep intramuscular injection into the gluteal muscle, the only site for which clinical experience has been obtained [1][3]. SA Health has moved long-acting injections to the ventrogluteal site because it is safer than other sites and absorbs the medicine better [4]; it has substantial muscle bulk and lies away from the sciatic nerve and the superior gluteal vessels [5]. Oil-based testosterone undecanoate carries an uncommon risk of POME, usually within minutes of injection; slow injection, correct depth and a period of observation are the safeguards [1][6][7].

Anatomy and landmarking

The site overlies gluteus medius and minimus, bounded by the greater trochanter (lateral), the anterior superior iliac spine (anterior) and the iliac crest (posterior). Landmark as described in Section 1 [4][5]. Where landmarks are hard to feel, take time, ask a colleague to confirm, and choose the longer needle so the injection reaches muscle [5]. Appendix A: the landmarking figure in the SA Health brochure [4].

Needle and equipment selection

Draw up with an 18G needle, as the oil is viscous. Inject with a 20–22G needle [10] — this procedure specifies 21G, 38 mm for most adults and 50 mm where the BMI is over 30 [5]. A wider lumen lets the oil flow with less pressure.

Injection technique

Follow the procedure in Section 1. The points most often done poorly are injecting too fast (the target is about 2 minutes for the 4 mL [10]), injecting too shallow, and massaging the site afterwards; each increases the chance of oil entering the circulation or leaking back along the needle track [5][6][7].

Recognising and managing complications

- **POME:** sudden cough, urge to cough, breathlessness, chest tightness, sweating, dizziness, paraesthesia or syncope within minutes of injection. Manage as in Section 1. Most episodes are self-limiting; escalate if severe [1][6][7].
- **Anaphylaxis:** manage per ASCIA — lay flat, adrenaline 1:1000 IM outer mid-thigh, 0.01 mg/kg up to 0.5 mg (adult), call 000, repeat after 5 minutes if needed, observe at least 4 hours after the last dose [8].
- **Local:** pain, haematoma, oil nodule, infection — manage per local wound care and escalate if infection is suspected [5].

Practical training and competency

- A short teaching session covering the Product Information, POME and anaphylaxis (30–45 minutes) [1][6][7][8].
- Demonstration of landmarking and injection on a model (15–20 minutes) [4][5].
- At least three supervised ventrogluteal injections, documented [5].
- Direct observation against the competency checklist (Appendix B), and a short quiz on recognising POME and anaphylaxis.

Patient information and counselling

- Explain the schedule — usually every 10 to 14 weeks once established [1][3].
- Explain that the injection goes into the side of the hip, takes about two minutes to give, and is followed by about 15 minutes of observation [10].
- Ask the patient to report sudden cough, breathlessness, dizziness, faintness or chest symptoms immediately, during the visit or after it [1][3].
- Patients should keep their Consumer Medicine Information leaflet [1].

Appendices

- Appendix A: landmarking figure, SA Health brochure [4].
- Appendix B: Competency assessment checklist (separate document).
- Appendix C: Adverse event reporting and escalation form (separate document).

References

[1] Aspen Pharmacare Australia. Reandron 1000 (testosterone undecanoate 1000 mg/4 mL) solution for injection — Australian Product Information and Consumer Medicine Information. Accessed via the Medicine Finder, Australian Commission on Safety and Quality in Health Care, 1 October 2026.

<https://www.safetyandquality.gov.au/medicine-finder/reandron-1000-solution-for-injection>

[2] Therapeutic Goods Administration. REANDRON 1000 testosterone undecanoate 1000 mg/4 mL injection vial — ARTG 213942. Canberra: TGA. <https://www.tga.gov.au/resources/artg/213942>

[3] Bayer New Zealand. Reandron 1000 (1000 mg/4 mL solution for injection) — New Zealand Data Sheet. Wellington: Medsafe. <https://medsafe.govt.nz/profs/Datasheet/r/Reandron1000inj.pdf>

[4] SA Health. Changes to your long acting injection — patient brochure on the ventrogluteal site (includes the landmarking figure). Adelaide: Department for Health and Wellbeing, Government of South Australia, 2022. [https://www.sahealth.sa.gov.au/wps/wcm/connect/e185e690-bc60-402f-b264-5ed189fd0390/Long+Acting+Intramuscular+Injections+Brochure+\(1\).pdf?MOD=AJPERES](https://www.sahealth.sa.gov.au/wps/wcm/connect/e185e690-bc60-402f-b264-5ed189fd0390/Long+Acting+Intramuscular+Injections+Brochure+(1).pdf?MOD=AJPERES)

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https://www.allergy.org.au/images/ASCIA_HP_Guidelines_Acute_Management_Anaphylaxis_July_2026.pdf
- [9] Therapeutic Goods Administration. Reporting adverse events — medicines and vaccines. Canberra: TGA.
<https://www.tga.gov.au/reporting-problems>
- [10] Laboratorios Liconsa. Guidance on the administration of testosterone undecanoate 1000 mg/4 mL — risk minimisation material approved by the MHRA, version 2.0, July 2024. (Same formulation as Reandron 1000.) <https://www.medicines.org.uk/emc/rmm/100105/Document>

Competency sign-off

Completed by	Date	Signature